Spatial Analysis of Cancer Mortality in Illinois (2020) Using R

1. Introduction

Cancer is a leading cause of mortality across the United States. Understanding geographic disparities in cancer mortality supports public health planning, resource allocation, and targeted policy interventions. This project examines the spatial distribution of cancer mortality rates across Illinois counties using areal data analysis in R. The analysis includes mapping, spatial autocorrelation statistics, hotspot detection via LISA, and assessing how poverty relates to cancer mortality patterns.

2. Data and Methods

2.1 Data Sources

Three core datasets were used:

- Cancer Mortality Data (CDC WONDER, 2020) - county-level cancer deaths
- County Population (ACS 2020) - used to compute mortality rates
- Percent in Poverty (ACS 2020) - socioeconomic covariate
- County Boundaries - TIGRIS shapefile of Illinois counties

Mortality rate was calculated as:

$$\text{Mortality Rate per 100,000} = \frac{\text{Deaths}}{\text{Population}} \times 100{,}000$$

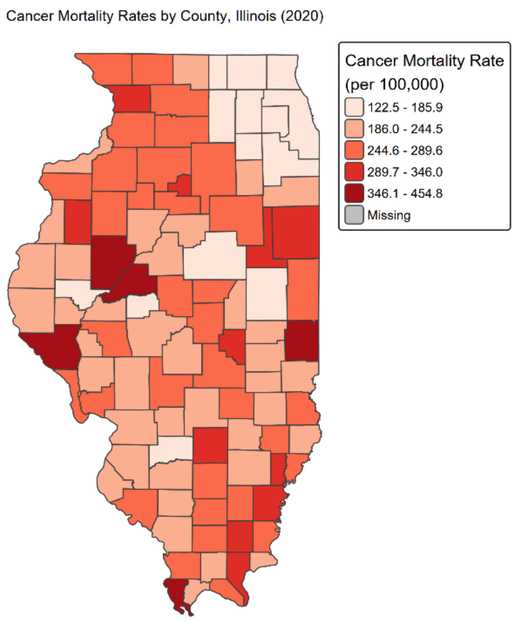
2.2 Data Processing in R

Analysis was performed using:

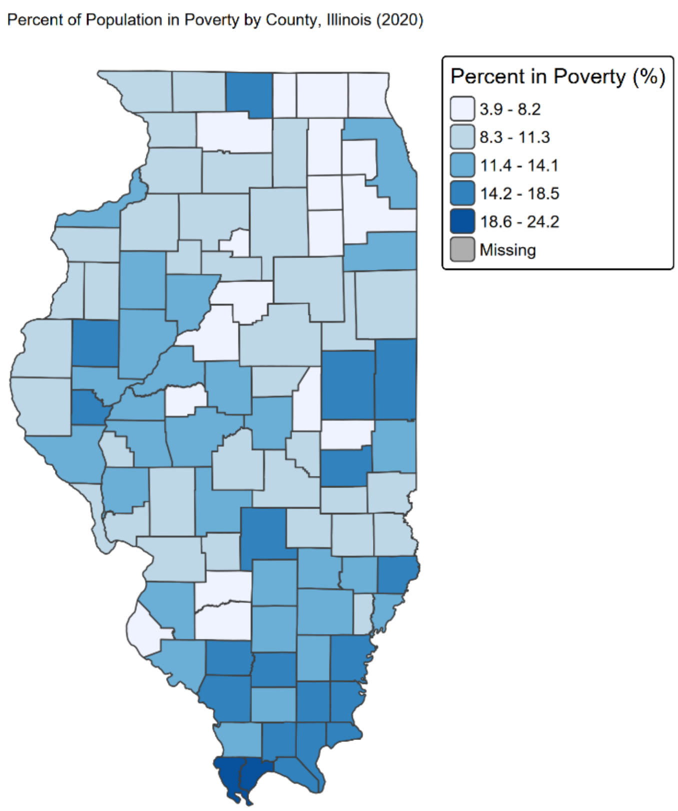
- sf for spatial data
- spdep for spatial weights & Moran's I
- tmap for thematic mapping
- tidyverse for data cleaning

Two main maps were created:

- Choropleth map of cancer mortality rates



- **Poverty rate map**



Both maps were saved as high-resolution PNG files.

2.3 Spatial Autocorrelation Analysis

Spatial weights were constructed using queen contiguity based on county boundaries.

Two levels of spatial autocorrelation were assessed:

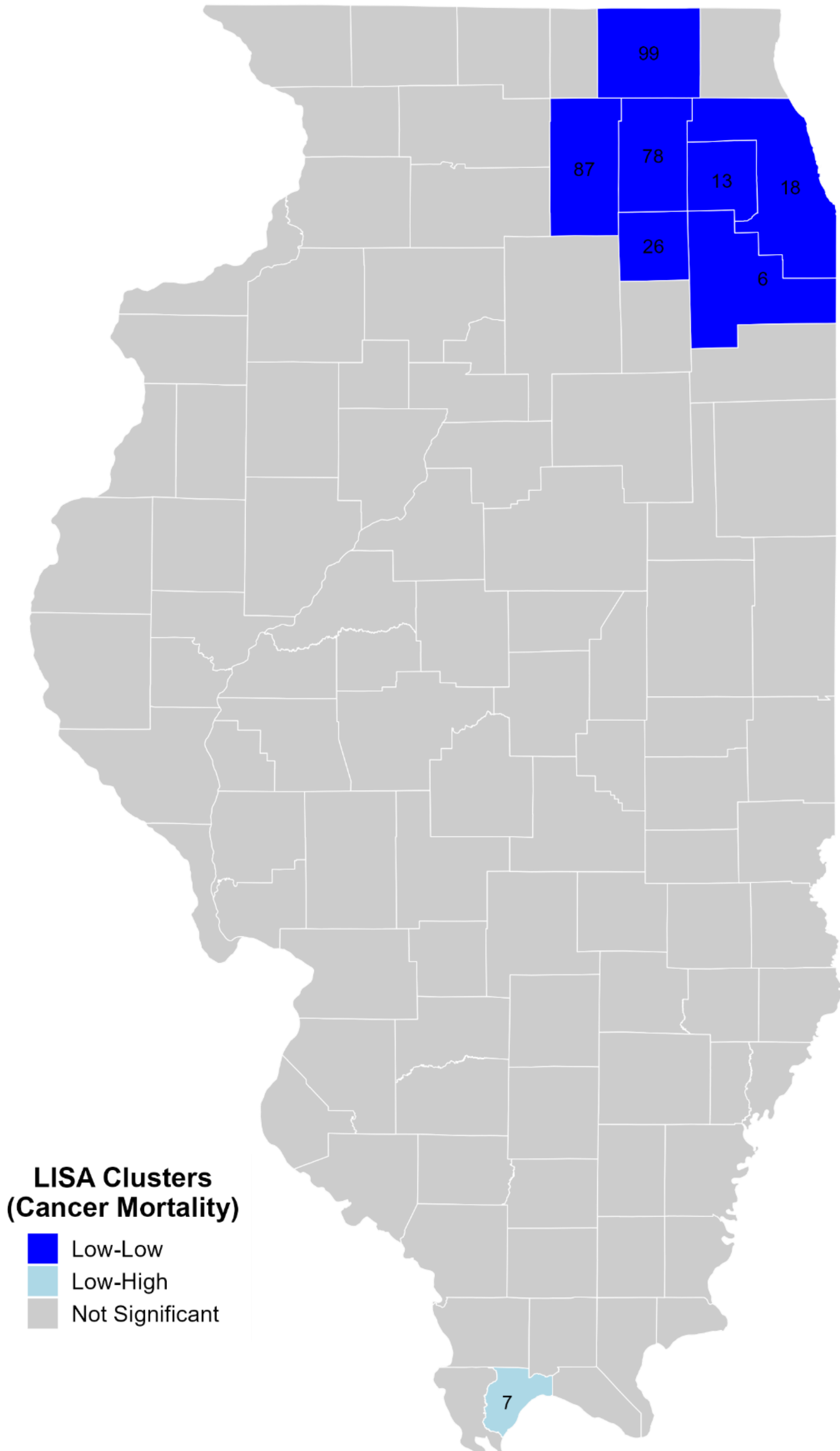
Global Moran's I

Evaluates whether mortality rates exhibit overall clustering.

Local Moran's I (LISA)

Detects localized hotspots, coldspots, and spatial outliers:

- High-High
- Low-Low
- High-Low
- Low-High

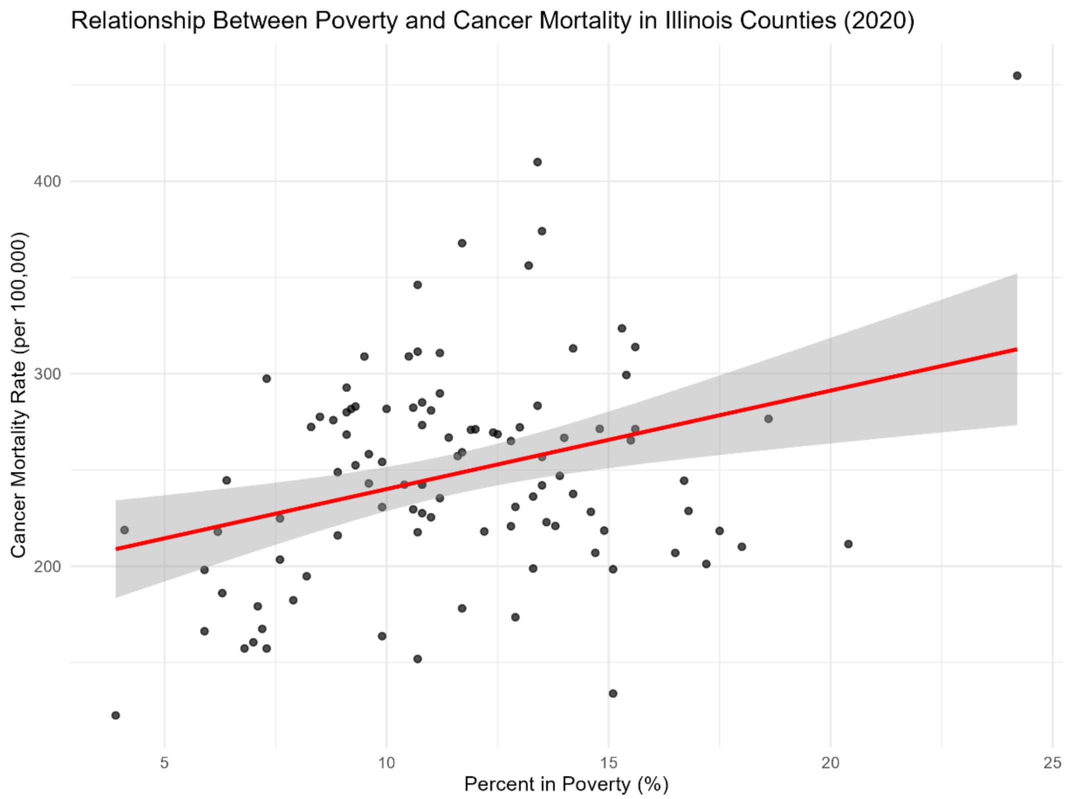

99
87
78
13
18
26
6
7
LISA Clusters
(Cancer Mortality)
Low-Low
Low-High
Not Significant

2.4 Poverty and Cancer Mortality Relationship

Two analyses were conducted:

1. **Pearson correlation**
2. **Simple linear regression**

A scatterplot was produced to visualize the association:



3. Results

3.1 Descriptive Statistics

Cancer Mortality Rate (per 100,000)

Statistic	Value
Minimum	122.48
1st Quartile	217.70

Statistic	Value
Median	244.50
Mean	248.10
3rd Quartile	277.30
Maximum	454.80

Percent in Poverty

Statistic	Value
Minimum	3.9
Median	11.2
Mean	11.58
Maximum	24.2

3.2 Global Moran's I

Global Moran's I for cancer mortality:

- **I = 0.1787**
- **Expected I = -0.0099**
- **Variance = 0.00371**
- **Z = 3.0948**
- **p = 0.0009847**

Interpretation:
Illinois exhibits significant positive spatial autocorrelation, meaning nearby counties tend to have similar cancer mortality rates.

3.3 Local Moran's I (LISA) Cluster Results

Eight counties exhibited significant local spatial structure.

Low–Low Coldspots (7 counties)

These counties have low cancer mortality surrounded by similar low-mortality counties.
All are located in northeastern Illinois, forming a large coldspot around the Chicago metropolitan area.

Low-High Outlier (1 county)

One county has low mortality but is surrounded by higher-mortality neighbors.

Table 1. Significant LISA Counties

County ID	County Name	Mortality Rate	Cluster Type
6	Will	167.41	Low-Low
13	DuPage	166.22	Low-Low
18	Cook	173.50	Low-Low
26	Kendall	122.48	Low-Low
78	Kane	157.25	Low-Low
87	DeKalb	163.65	Low-Low
99	McHenry	186.02	Low-Low
7	Pulaski	211.50	Low-High

3.4 Poverty Patterns

Poverty rates are highest in the southern and southeastern counties, with lower rates concentrated in northern Illinois.

The poverty map visually aligns with mortality patterns, but not perfectly, suggesting socioeconomic factors matter but are not the only drivers.

3.5 Correlation and Regression Results

Correlation

- $r = 0.3196$
- $p = 0.00106$

Interpretation:
There is a moderate, statistically significant positive correlation between poverty and cancer mortality.

Regression

$$\text{Mortality Rate} = 188.89 + 5.12(\%\text{Poverty})$$

- Slope p-value = 0.00106
- $R^2 = 0.1021$

Interpretation:
Every 1% increase in poverty corresponds to an average increase of 5.12 deaths per 100,000.

The scatterplot confirms this trend visually.

4. Discussion

Spatial Patterns

The northeastern cluster (Counties 6, 13, 18, 26, 78, 87, 99) forms a large Low-Low coldspot. These counties have:

- higher socioeconomic conditions
- better access to healthcare
- urban/suburban advantages

Pulaski County (ID 7) stands out as a Low–High outlier, meaning its mortality is low but surrounded by higher-mortality neighbors — reflecting spatial inequality in the southern region.

Poverty and Mortality

The significant correlation and regression results indicate that poverty is an important factor, though not the only one. Other potential contributors include:

- rural healthcare access
- environmental exposures
- differences in cancer screening
- age structure
- behavioral factors

5. Conclusion

This spatial analysis revealed:

1. Cancer mortality in Illinois displays significant spatial clustering.
2. A major Low-Low coldspot exists in northeastern Illinois.

3. One Low-High outlier was detected in Pulaski County.
4. Poverty has a significant positive relationship with cancer mortality.
5. Socioeconomic and geographic inequality likely drive these disparities.

The analysis demonstrates the value of combining spatial statistics and regression to understand public health outcomes.

References

- Anselin, Luc. "Local indicators of spatial association—LISA." *Geographical analysis* 27, no. 2 (1995): 93-115.
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- Illinois Department of Public Health. http://dph.illinois.gov/data-statistics/vital-statistics/death-statistics.html
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- Wang, Fahui. "Spatial clusters of cancers in Illinois 1986–2000." *Journal of Medical Systems* 28, no. 3 (2004): 237-256.